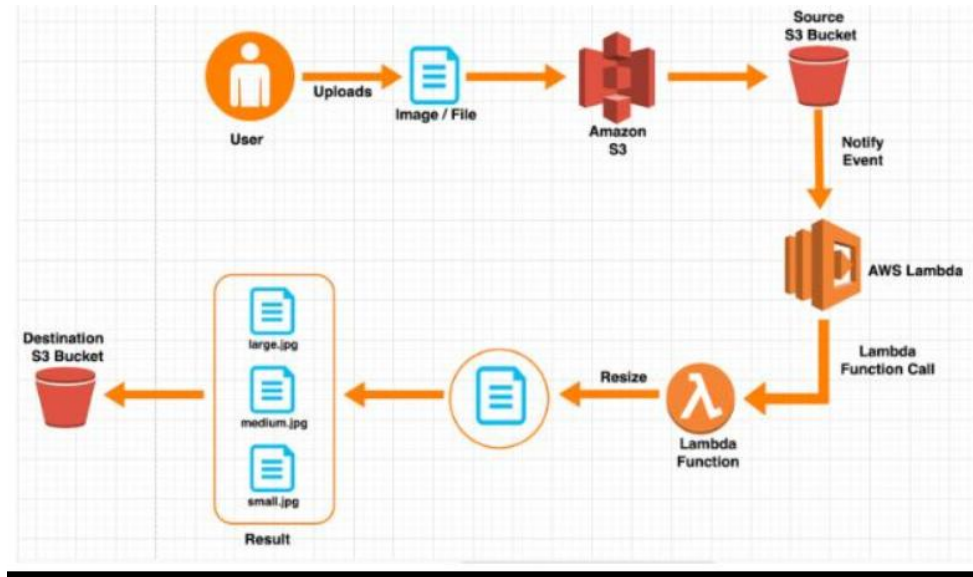


Name : Vivek Chaudhari

Project : 9

ImgScale : Automatic image resizing using serverless architecture



Project Overview

ImgScale is a serverless image processing system built on AWS that automatically resizes images when they are uploaded to cloud storage. It eliminates the need for servers by using event-driven computing, ensuring scalability, low cost, and real-time processing. **Working Flow (End-to-End)**

1. User uploads an image to a storage bucket
2. Storage service triggers a serverless function
3. The function processes the image (resize/compress)
4. The resized image is stored in another bucket (or folder)
5. User can access the optimized image

⚙️ AWS Services Used & Their Role

📦 Amazon S3

- Stores original and processed images
- Triggers events when a new file is uploaded
- Highly durable and scalable storage

AWS Lambda

- Executes code automatically on image upload
- Resizes images using Python libraries (like Pillow)
- No server management required

Core Concept: Event-Driven Architecture

- Action (image upload) → triggers function
- No manual execution needed
- Fully automated pipeline

Key Objectives Achieved

- ✓ Fully automated image processing
- ✓ Serverless (no infrastructure management)
- ✓ Scalable for high uploads
- ✓ Cost-efficient (pay per execution)

Real-World Use Cases

- Social media apps (image optimization)
- E-commerce (product image thumbnails)
- Content delivery systems

Amazon S3

> Buckets

Amazon S3

> Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (2)

Info

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

Name

AWS Region

Creation date

input-images-hitesh

Asia Pacific (Mumbai) ap-south-1

April 24, 2026, 23:36:17 (UTC+05:30)

output-images-hitesh

Asia Pacific (Mumbai) ap-south-1

April 24, 2026, 23:36:54 (UTC+05:30)

Account snapshot

Info

Updated daily

View dashboard

Storage Lens provides visibility into storage usage and activity trends.

External access summary

Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

Amazon S3

> Buckets

> input-images-hitesh

> uploads/

Amazon S3

> Buckets

> input-images-hitesh

> uploads/

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uploads/

Copy S3 URI

Objects

Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name

Type

Last modified

Size

download.jpeg

jpeg

April 25, 2026, 15:07:44 (UTC+05:30)

16.0 KB

Amazon S3

> Buckets

> input-images-hitesh

> uploads/

> download.jpeg

Amazon S3

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> input-images-hitesh

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download.jpeg

Info

Copy S3 URI

Download

Open

Object actions

Properties

Permissions

Versions

Object overview

Owner

bf438702eda18f8d7bb27d6fbde9b7041004e0be709133843a85302c28ff3502

AWS Region

Asia Pacific (Mumbai) ap-south-1

Last modified

April 25, 2026, 15:07:44 (UTC+05:30)

Size

16.0 KB

Type

jpeg

Key

uploads/download.jpeg

S3 URI

s3://input-images-hitesh/uploads/download.jpeg

Amazon Resource Name (ARN)

arn:aws:s3::input-images-hitesh/uploads/download.jpeg

Entity tag (Etag)

85b9be415723c7e8ed6ce344fe709ca7

Object URL

<https://input-images-hitesh.s3.ap-south-1.amazonaws.com/uploads/download.jpeg>

Amazon S3

> Buckets

> output-images-hitesh

> uploads/

Amazon S3

> Buckets

> output-images-hitesh

> uploads/

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Find objects by prefix

< 1 >

Name

Type

Last modified

Size

Storage class

download.jpeg

jpeg

April 25, 2026, 15:07:47 (UTC+05:30)

13.7 KB

Standard

Amazon S3 > Buckets > output-images-hitesh > uploads/ > download.jpeg

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations

download.jpeg

Info

[Copy S3 URI](#) [Download](#) [Open ↗](#) [Object actions](#)

Properties

Object overview

Owner
bf438702eda18f8d7bb27d6fbd9b7041004e0be709133843a85302c28ff3502

AWS Region
Asia Pacific (Mumbai) ap-south-1

Last modified
April 25, 2026, 15:07:47 (UTC+05:30)

Size
13.7 KB

Type
jpeg

Key
[uploads/download.jpeg](#)

S3 URI
[s3://output-images-hitesh/uploads/download.jpeg](#)

Amazon Resource Name (ARN)
[arn:aws:s3::output-images-hitesh/uploads/download.jpeg](#)

Entity tag (Etag)
[a6d7aa0949bc5275b430fc933ad141e0](#)

Object URL
[https://output-images-hitesh.s3.ap-south-1.amazonaws.com/uploads/download.jpeg](#)

CloudWatch > Log management > /aws/lambda/image-resizer > 2026/04/25/[\$LATEST]edef649ef49422fbda77b4e5e1e2a80

CloudWatch

- Favorites and recents
- Ingestion
- Dashboards
- Alarms [0](#) [0](#) [0](#) [0](#)
- ▶ AI Operations
- ▶ GenAI Observability
- ▶ Application Signals (APM)
- ▶ Infrastructure Monitoring
- ▼ Logs
 - [Log Management](#)
 - Log Analytics [Preview](#)
 - Log Anomalies
 - Live Tail
 - Logs Insights

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

[Clear](#) [1m](#) [30m](#) [1h](#) [12h](#) [Custom](#) [UTC timezone](#)

[Display](#)

Timestamp	Message
	No older events at this moment. Retry
2026-04-25T09:37:44.131Z	INIT_START Runtime Version: python:3.10.mainlinev2.v7 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:04c1f9d1f217da55ae2226b...
2026-04-25T09:37:44.826Z	START RequestId: acb24ef4-4180-4dd5-8aca-6779e69dcb79 Version: \$LATEST
2026-04-25T09:37:44.826Z	File: uploads/download.jpeg
2026-04-25T09:37:46.631Z	END RequestId: acb24ef4-4180-4dd5-8aca-6779e69dcb79
2026-04-25T09:37:46.631Z	REPORT RequestId: acb24ef4-4180-4dd5-8aca-6779e69dcb79 Duration: 1805.12 ms Billed Duration: 2497 ms Memory Size: 128 MB Max Memory U...
2026-04-25T09:39:41.749Z	START RequestId: 718809e0-52f3-4bd4-8796-4bdce3804865 Version: \$LATEST
2026-04-25T09:39:41.750Z	File: uploads/test.jpg
2026-04-25T09:39:44.917Z	END RequestId: 718809e0-52f3-4bd4-8796-4bdce3804865
2026-04-25T09:39:44.917Z	REPORT RequestId: 718809e0-52f3-4bd4-8796-4bdce3804865 Duration: 3000.00 ms Billed Duration: 3000 ms Memory Size: 128 MB Max Memory U...

image-resizer

Function overview

Image-resizer

Layers

[Add integrator](#)

Function ARN

arn:aws:lambda:ap-south-1:8374115405775:function:image-resizer

Description

Last modified

30/03/2026 (UTC)

[Export to Infrastructure Composer](#) [Download](#)

[Test](#) [Monitor](#) [Configuration](#) [Aliases](#) [Versions](#)

Code source

image-resizer

```
1 function handler(event, context) {
2   console.log('Received event: ' + JSON.stringify(event, null, 2));
3   const s3 = new AWS.S3({
4     accessKeyId: process.env.ACCESS_KEY_ID,
5     secretAccessKey: process.env.SECRET_ACCESS_KEY,
6     region: process.env.AWS_REGION
7   });
8   const bucketName = event['Records'][0]['s3']['bucket']['name'];
9   const key = event['Records'][0]['s3']['object']['key'];
10  console.log('Bucket: ' + bucketName + ', Key: ' + key);
11
12  // Get the object from S3
13  const response = s3.getObject({Bucket: bucketName, Key: key});
14  const imageBuffer = response.Body;
15  const image = imageBuffer.toArrayBuffer();
16
17  // Resize the image
18  const imageResized = image.resize(200, 200);
19
20  // Save the resized image to S3
21  const outputBuffer = imageResized.toArrayBuffer();
22  const outputKey = 'output-' + key;
23
24  // Upload the resized image to S3
25  s3.putObject({Bucket: bucketName, Key: outputKey, Body: outputBuffer});
26
27  console.log('Resized image uploaded to S3');
28
29  return {statusCode: 200, body: 'Image resized successfully'};
30}
```